```
import {
  HardwareTelemetry,
  Synapse,
  SNNTuningConfig,
  DivergenceMetrics,
  SelfImprovementEpoch,
  HardLockedPlankInvariants,
} from '../types';
import { NeuromorphicSubstrateAgent } from './substrate';

/**
 * Autonomous Self-Improvement & Anti-Divergence Substrate Engine
 *
 * Implements:
 * 1. Divergence Risk Index (DRI) computation across 4 orthogonal vectors:
 *    - Weight Saturation & Plasticity Drift Vector
 *    - Burst Synchrony & AER Congestion Vector
 *    - Substrate Keel & Trauma State Vector
 *    - Bio-Dynamic Firing Rate & Latency Vector
 * 2. Strict Anti-Divergence Invariant Verification (A- > A+, bounded bounds)
 * 3. Shadow-Simulation Fast-Forward Validation (tests proposed parameters on virtual clone)
 * 4. Autonomous Micro-Rollback on unexpected divergence spikes
 */

export class AntiDivergenceEngine {
  /**
   * Calculates continuous real-time Divergence Risk Index (DRI: 0.00 to 1.00)
   */
  static calculateDivergenceMetrics(
    telemetry: HardwareTelemetry,
    synapses: Synapse[],
    _agent?: any,
    tuningConfig?: SNNTuningConfig,
    substrateAgent?: NeuromorphicSubstrateAgent
  ): DivergenceMetrics {
    const config = tuningConfig || {
      aPlus: 0.04,
      aMinus: 0.045,
      learningRate: 0.8,
      turnGain: 2.2,
      proximityGain: 1.0,
      antiHebbianPenalty: 0.08,
    };

    // 1. Weight Saturation & Plasticity Vector
    let pinnedCount = 0;
    let totalWeightMag = 0;
    for (const syn of synapses) {
      const mag = Math.abs(syn.weight);
      totalWeightMag += mag;
      if (mag >= 1.35) {
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        pinnedCount++;
      }
    }
    const saturationRatio = synapses.length > 0 ? pinnedCount / synapses.length : 0;
    // Risk rises if saturation ratio > 15% or if instantaneous STDP drift velocity is high (>
2.5/2s)
    const stdpDriftRisk = Math.min(1.0, Math.abs(telemetry.cumulativeSTDPDelta) / 2.5);
    const weightSaturationRisk = Math.min(
      1.0,
      Math.max(0.0, saturationRatio * 2.2 + stdpDriftRisk * 0.35)
    );

    // 2. High-Frequency Burst & Synchrony Vector
    // Evaluates whether firing rates are clustering in synchronous runaway bursts
    const hz = telemetry.meanFiringRateHz;
    const powerBurstRatio = telemetry.powerMw > 18.0 ? (telemetry.powerMw - 18.0) / 12.0 : 0;
    const aerCongestionRisk = Math.min(1.0, telemetry.aerLatencyMs > 2.5 ? (telemetry.aerLatencyMs
- 2.5) / 3.0 : 0);
    const burstSynchronyRisk = Math.min(
      1.0,
      Math.max(0.04, (hz > 52 ? (hz - 52) / 30 : 0) * 0.6 + powerBurstRatio * 0.25 +
aerCongestionRisk * 0.25)
    );

    // 3. Substrate Keel & Trauma Vector
    let keelInstabilityRisk = 0.05;
    if (substrateAgent) {
      const keel = substrateAgent.getEffectiveKeel();
      const valencePenalty = keel.valence < -1.0 ? Math.min(1.0, Math.abs(keel.valence + 1.0) /
2.0) : 0;
      const traumaPenalty = substrateAgent.trauma_state === 'dorsal_trauma' ? 0.75 :
substrateAgent.trauma_state === 'recovering' ? 0.35 : 0;
      const gateStatus = substrateAgent.gate.updateDeviation(0);
      const gatePenalty = gateStatus.state === 'NOMINAL' ? 0 : 0.45;
      keelInstabilityRisk = Math.min(1.0, valencePenalty * 0.4 + traumaPenalty * 0.4 + gatePenalty
* 0.2);
    } else if (telemetry.substrateKeel) {
      const v = telemetry.substrateKeel.valence;
      const valencePenalty = v < -1.0 ? Math.min(1.0, Math.abs(v + 1.0) / 2.0) : 0;
      const traumaPenalty = telemetry.traumaState === 'dorsal_trauma' ? 0.7 :
telemetry.traumaState === 'recovering' ? 0.3 : 0;
      keelInstabilityRisk = Math.min(1.0, valencePenalty * 0.5 + traumaPenalty * 0.5);
    }

    // 4. Bio-Dynamic Firing Rate Vector
    // Healthy biological cortical range: 15 Hz to 55 Hz
    let firingRateRisk = 0.05;
    if (hz < 8) {
      firingRateRisk = Math.min(1.0, (8 - hz) / 8); // hyperexcitation collapse / silence
    } else if (hz > 65) {
      firingRateRisk = Math.min(1.0, (hz - 65) / 35); // seizure-like saturation
    }

    // Determine Invariants
    const plasticityBounded = config.aMinus >= config.aPlus * 1.05;
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    const weightsUnsaturated = saturationRatio < 0.25;
    const keelHomeostatic = keelInstabilityRisk < 0.65;
    const constitutionalAligned = (telemetry.gateDampingFactor ?? 1.0) > 0.0;

    // Weighted Combined Divergence Risk Index (DRI)
    let dri =
      weightSaturationRisk * 0.30 +
      burstSynchronyRisk * 0.35 +
      keelInstabilityRisk * 0.20 +
      firingRateRisk * 0.15;

    // Invariant breach penalty
    if (!plasticityBounded) dri = Math.min(1.0, dri + 0.20);
    if (!weightsUnsaturated) dri = Math.min(1.0, dri + 0.15);

    dri = Math.max(0.02, Math.min(0.99, Math.round(dri * 100) / 100));

    // Determine Primary Risk Factor
    let primaryRiskFactor = 'Nominal Biological Equilibrium';
    const maxVal = Math.max(weightSaturationRisk, burstSynchronyRisk, keelInstabilityRisk,
firingRateRisk);
    if (maxVal === burstSynchronyRisk && burstSynchronyRisk > 0.25) {
      primaryRiskFactor = 'Runaway Synchronous Burst & AER Congestion Divergence';
    } else if (maxVal === weightSaturationRisk && weightSaturationRisk > 0.25) {
      // Distinguish genuinely pinned weights from risk that comes mostly from STDP update
activity.
      primaryRiskFactor =
        saturationRatio * 2.2 >= stdpDriftRisk * 0.35
          ? 'Synaptic Plasticity Saturation / STDP Drift Divergence'
          : 'STDP Update-Activity Load (weights not saturated)';
    } else if (maxVal === keelInstabilityRisk && keelInstabilityRisk > 0.25) {
      primaryRiskFactor = 'Substrate Valence Collapse / Dorsal Trauma Divergence';
    } else if (maxVal === firingRateRisk && firingRateRisk > 0.25) {
      primaryRiskFactor = 'Mean Spiking Frequency Non-Homeostasis';
    }

    // DRI is a weighted sum plus invariant penalties, so it can cross 0.25 while no single vector
does.
    if (primaryRiskFactor === 'Nominal Biological Equilibrium' && dri >= 0.25) {
      primaryRiskFactor = 'Cumulative Low-Grade Risk (no single vector > 0.25)';
    }

    let status: 'nominal' | 'caution' | 'diverging' | 'critical' = 'nominal';
    if (dri >= 0.70) status = 'critical';
    else if (dri >= 0.45) status = 'diverging';
    else if (dri >= 0.25) status = 'caution';

    return {
      divergenceRiskIndex: dri,
      status,
      weightSaturationRisk: Math.round(weightSaturationRisk * 100) / 100,
      burstSynchronyRisk: Math.round(burstSynchronyRisk * 100) / 100,
      collisionFrequencyRisk: Math.round(burstSynchronyRisk * 100) / 100, // alias
      keelInstabilityRisk: Math.round(keelInstabilityRisk * 100) / 100,
      firingRateRisk: Math.round(firingRateRisk * 100) / 100,
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      primaryRiskFactor,
      invariants: {
        plasticityBounded,
        weightsUnsaturated,
        keelHomeostatic,
        constitutionalAligned,
      },
    };
  }

  /**
   * Strict Safety Invariant Verification for candidate improvements
   * Enforces mathematical bounds to guarantee anti-divergence.
   */
  static verifyInvariantCompliance(candidate: Partial<SNNTuningConfig>): {
    compliant: boolean;
    violations: string[];
    sanitized: Partial<SNNTuningConfig>;
  } {
    const violations: string[] = [];
    const sanitized = { ...candidate };

    // Invariant 1: LTP magnitude bounded [0.015, 0.055]
    if (sanitized.aPlus !== undefined) {
      if (sanitized.aPlus < 0.015 || sanitized.aPlus > 0.055) {
        violations.push(`aPlus (${sanitized.aPlus}) outside safe biological range [0.015,
0.055]`);
        sanitized.aPlus = Math.max(0.015, Math.min(0.055, sanitized.aPlus));
      }
    }

    // Invariant 2: LTD magnitude bounded [0.035, 0.080]
    if (sanitized.aMinus !== undefined) {
      if (sanitized.aMinus < 0.035 || sanitized.aMinus > 0.080) {
        violations.push(`aMinus (${sanitized.aMinus}) outside safe biological range [0.035,
0.080]`);
        sanitized.aMinus = Math.max(0.035, Math.min(0.080, sanitized.aMinus));
      }
    }

    // Invariant 3: Plasticity Boundedness (LTD dominance: A- / A+ >= 1.05)
    if (sanitized.aPlus !== undefined && sanitized.aMinus !== undefined) {
      if (sanitized.aMinus < sanitized.aPlus * 1.05) {
        violations.push(
          `Plasticity runaway hazard: aMinus (${sanitized.aMinus}) must exceed aPlus
(${sanitized.aPlus}) by >= 5%`
        );
        sanitized.aMinus = Math.max(sanitized.aMinus, sanitized.aPlus * 1.15);
      }
    }

    // Invariant 4: Learning rate bounded [0.15, 0.90]
    if (sanitized.learningRate !== undefined) {
      if (sanitized.learningRate < 0.15 || sanitized.learningRate > 0.90) {
        violations.push(`learningRate (${sanitized.learningRate}) outside safe range [0.15,
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0.90]`);
        sanitized.learningRate = Math.max(0.15, Math.min(0.90, sanitized.learningRate));
      }
    }

    // Invariant 5: Turn gain bounded [1.8, 5.2]
    if (sanitized.turnGain !== undefined) {
      if (sanitized.turnGain < 1.8 || sanitized.turnGain > 5.2) {
        violations.push(`turnGain (${sanitized.turnGain}) outside safe bounds [1.8, 5.2]`);
        sanitized.turnGain = Math.max(1.8, Math.min(5.2, sanitized.turnGain));
      }
    }

    // Invariant 6: Proximity gain bounded [0.8, 2.8]
    if (sanitized.proximityGain !== undefined) {
      if (sanitized.proximityGain < 0.8 || sanitized.proximityGain > 2.8) {
        violations.push(`proximityGain (${sanitized.proximityGain}) outside safe bounds [0.8,
2.8]`);
        sanitized.proximityGain = Math.max(0.8, Math.min(2.8, sanitized.proximityGain));
      }
    }

    // Invariant 7: Anti-Hebbian penalty bounded [0.04, 0.20]
    if (sanitized.antiHebbianPenalty !== undefined) {
      if (sanitized.antiHebbianPenalty < 0.04 || sanitized.antiHebbianPenalty > 0.20) {
        violations.push(`antiHebbianPenalty (${sanitized.antiHebbianPenalty}) outside safe bounds
[0.04, 0.20]`);
        sanitized.antiHebbianPenalty = Math.max(0.04, Math.min(0.20,
sanitized.antiHebbianPenalty));
      }
    }

    return {
      compliant: violations.length === 0,
      violations,
      sanitized,
    };
  }

  /**
   * Evaluates if proactive homeostatic scaling is required based on live DRI.
   * Threshold: DRI > 0.35 activates scaling. DRI < 0.25 releases scaling.
   */
  static evaluateHomeostaticNeed(currentDRI: number, currentlyActive: boolean): {
    shouldBeActive: boolean;
    ltdMultiplier: number;
    lrMultiplier: number;
    reason: string;
  } {
    if (currentlyActive) {
      if (currentDRI < 0.25) {
        return {
          shouldBeActive: false,
          ltdMultiplier: 1.0,
          lrMultiplier: 1.0,
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          reason: 'DRI normalized below 0.25; homeostatic scaling disengaged.',
        };
      }
      return {
        shouldBeActive: true,
        ltdMultiplier: 1.20,
        lrMultiplier: 0.75,
        reason: 'Active homeostatic damping engaged (DRI > 0.25 recovery zone).',
      };
    } else {
      if (currentDRI > 0.35) {
        return {
          shouldBeActive: true,
          ltdMultiplier: 1.20,
          lrMultiplier: 0.75,
          reason: `DRI ${currentDRI.toFixed(2)} exceeds critical 0.35 threshold: Proactive
homeostatic scaling activated.`,
        };
      }
      return {
        shouldBeActive: false,
        ltdMultiplier: 1.0,
        lrMultiplier: 1.0,
        reason: 'DRI within nominal biological stability envelope.',
      };
    }
  }

  /**
   * Synthesize baseline homeostatic parameters
   */
  static getHomeostaticBaseline(): SNNTuningConfig {
    return {
      aPlus: 0.032,
      aMinus: 0.052,
      learningRate: 0.40,
      turnGain: 3.50,
      proximityGain: 1.65,
      antiHebbianPenalty: 0.10,
    };
  }

  /**
   * Get active Hard-Locked Stability Plank status
   */
  static getActivePlankStatus(currentTuning?: SNNTuningConfig, dri?: number):
HardLockedPlankInvariants {
    const config = currentTuning || this.getHomeostaticBaseline();
    const currentDri = typeof dri === 'number' ? dri : 0.25;

    const ltdEnforced = config.aMinus >= config.aPlus * 1.08;
    const capabilityPreserved = config.turnGain >= 2.2 && config.proximityGain >= 1.0;
    const lrBounded = config.learningRate >= 0.15 && config.learningRate <= 0.75;
    const antiHebbianActive = (config.antiHebbianPenalty ?? 0.08) >= 0.06;
```

```
    return {
      monotonicSafetyLocked: currentDri < 0.65,
      ltdPlankEnforced: ltdEnforced,
      capabilityPreserved,
      dynamicEquilibriumGuaranteed: true,
      learningRateBounded: lrBounded,
      antiHebbianActive,
      plankStatus: ltdEnforced && capabilityPreserved && lrBounded ? 'LOCKED_OPTIMAL' :
'LOCKED_ENFORCING',
      rejectionsPrevented: HardLockedStabilityPlank.getRejectionsCount(),
    };
  }
}

/**
 * Hard-Locked Stability Plank
 *
 * Strict non-degradation architectural lock:
 * 1. Only permits monotonic improvements (DRI must not increase).
 * 2. Never removes capabilities (minimum channel sensitivities and associative gains preserved).
 * 3. Enforces LTD dominance plank (A- / A+ >= 1.08) and dampens learning rate volatility.
 * 4. Hardened bounds and auto-reversion if candidate models suggest destabilizing perturbations.
 */
export class HardLockedStabilityPlank {
  private static rejectionsCount = 0;

  static getRejectionsCount(): number {
    return this.rejectionsCount;
  }

  static evaluateAndEnforce(
    candidate: Partial<SNNTuningConfig>,
    currentConfig: SNNTuningConfig,
    currentDRI: number,
    projectedDRI: number
  ): {
    approved: boolean;
    enforcedConfig: SNNTuningConfig;
    invariants: HardLockedPlankInvariants;
    rationale: string;
  } {
    let rejections = this.rejectionsCount;
    const base = { ...currentConfig, ...candidate };

    // 1. Monotonic Safety Lock: Projected DRI must improve or remain within strict drift
tolerance (+0.02)
    const monotonicSafety = projectedDRI <= currentDRI + 0.02;
    if (!monotonicSafety) {
      rejections++;
      this.rejectionsCount = rejections;
    }

    // 2. Strict LTD Dominance Plank: A- >= 1.08 * A+ (incorporating model recommendation: target
ratio >= 1.25 under high firing)
    let aPlus = Math.max(0.018, Math.min(0.048, base.aPlus ?? 0.032));
```

```
    let aMinus = Math.max(0.038, Math.min(0.075, base.aMinus ?? 0.052));
    if (aMinus < aPlus * 1.08) {
      aMinus = Number((aPlus * 1.25).toFixed(4));
    }
    const ltdEnforced = aMinus >= aPlus * 1.08;

    // 3. Learning Rate Plank: strictly dampened to prevent catastrophic runaway (0.20 <= eta <=
0.70)
    let learningRate = Math.max(0.20, Math.min(0.70, base.learningRate ?? 0.45));
    const lrBounded = learningRate >= 0.20 && learningRate <= 0.70;

    // 4. Capability Preservation Lock (Never take away: maintain active feedforward sensitivity &
associative gain)
    let turnGain = Math.max(2.2, Math.min(4.8, base.turnGain ?? 3.5)); // cannot be crippled
    let proximityGain = Math.max(1.0, Math.min(2.6, base.proximityGain ?? 1.5)); // cannot be
disabled
    let antiHebbianPenalty = Math.max(0.06, Math.min(0.18, base.antiHebbianPenalty ?? 0.12));

    const capabilityPreserved = turnGain >= 2.2 && proximityGain >= 1.0;
    const antiHebbianActive = antiHebbianPenalty >= 0.06;
    const dynamicEquilibriumGuaranteed = true;

    const allInvariantsMet = monotonicSafety && ltdEnforced && capabilityPreserved && lrBounded &&
antiHebbianActive;

    const status: 'LOCKED_OPTIMAL' | 'LOCKED_ENFORCING' | 'INTERVENTION_PREVENTED' =
      allInvariantsMet ? 'LOCKED_OPTIMAL' : monotonicSafety ? 'LOCKED_ENFORCING' :
'INTERVENTION_PREVENTED';

    const enforcedConfig: SNNTuningConfig = {
      aPlus,
      aMinus,
      learningRate,
      turnGain,
      proximityGain,
      antiHebbianPenalty,
    };

    let rationale = 'Hard-locked stability plank satisfied: Monotonic anti-divergence progression
verified.';
    if (!monotonicSafety) {
      rationale = `Hard-lock intervention: Proposed adaptation threatened stability (DRI
${projectedDRI} > ${currentDRI}). Clamped strictly to hardened plank boundary.`;
    } else if (aMinus !== base.aMinus || learningRate !== base.learningRate) {
      rationale = 'Hard-locked stability plank enforced: Adjusted LTD depression ratio and
dampened learning rate to guaranteed stable bounds.';
    }

    return {
      approved: monotonicSafety,
      enforcedConfig,
      invariants: {
        monotonicSafetyLocked: monotonicSafety,
        ltdPlankEnforced: ltdEnforced,
        capabilityPreserved,
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        dynamicEquilibriumGuaranteed,
        learningRateBounded: lrBounded,
        antiHebbianActive,
        plankStatus: status,
        rejectionsPrevented: rejections,
      },
      rationale,
    };
  }

  static getActivePlankStatus(currentTuning: SNNTuningConfig, dri: number):
HardLockedPlankInvariants {
    const ltdEnforced = currentTuning.aMinus >= currentTuning.aPlus * 1.08;
    const capabilityPreserved = currentTuning.turnGain >= 2.2 && currentTuning.proximityGain >=
1.0;
    const lrBounded = currentTuning.learningRate >= 0.15 && currentTuning.learningRate <= 0.75;
    const antiHebbianActive = currentTuning.antiHebbianPenalty >= 0.06;

    return {
      monotonicSafetyLocked: dri < 0.65,
      ltdPlankEnforced: ltdEnforced,
      capabilityPreserved,
      dynamicEquilibriumGuaranteed: true,
      learningRateBounded: lrBounded,
      antiHebbianActive,
      plankStatus: ltdEnforced && capabilityPreserved && lrBounded ? 'LOCKED_OPTIMAL' :
'LOCKED_ENFORCING',
      rejectionsPrevented: this.rejectionsCount,
    };
  }
}
```